DEPARTMENT OF WORKS & HIGHWAYS
MATERIAL SCIENCE & TECHNOLOGY BRANCH
Materials Testing Laboratory

☑ Port Moresby ☐ Mt. Hagen ☐ Lae ☐ Annex Lab.

Date:
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PARTICLE SIZE DISTRIBUTION WORKSHEET - AS.1289.3.6.1

Client : Asset No.:
Project: Material Source :
Location: Pit No.:
Material Description: Depth:

COARSE SIEVING

	Mass of container + total wet sample (m1)		g		TEST PORTION FOR COARSE GRADING
	Mass of container + total dry sample (m2)		g		REQUIRED MINIMUM
	Mass of container (m3)		g		MASS OF (800 - 1000g)
	Mass of water (m1 - m2) = (m4)		g	**Calculation Details**	(200mm dia. Sieve)
	Total Dry mass of unwashed sample (m2 - m3) = (m5)		g		
MAXIMUM MASS	Moisture content (m4) / (m5) x =(w)		%	100/(m5) = FACTOR 1	
RETAINED EACH	Check : (m5)x(w +100)/(100)+(m3) = m1 or (m1 - m3) / (w+100)x(100) = m5				

SIEVE SIZES (mm)	AS.Sieve Size (mm)	Individual mass Retained (m6)	Individual percent Retained(m7)	Total percent Passing		
3000	75.00 mm				(F1) x (m6) = m7	MINIMUM REQUIRED TEST
2750	63.00 mm					PORTION FOR
2200	37.50 mm				F1 = _____	MASS OF (1800-3750g)
1800	26.50 mm					(300mm dia. Sieve)
	Pan 1					

INTERMEDIATE SIEVING

MAXIMUM MASS	Mass of dry subsample (m8)			g		MINIMUM REQUIRED TEST
RETAINED EACH	Mass of Adjusted subsample (M_8 x P26.5)/100 = (m9)			g	**Calculation Details**	PORTION FOR
SIEVE SIZES (mm)	AS.Sieve Size (mm)	Individual mass Retained (m10)	Individual percent Retained(m11)	Total percent Passing	100/(m9) = FACTOR 2	INTERMEDIATE GRADING
1200	19.00 mm				(F2) x (m10) = m11	MASS OF (400 - 600g)
900	13.20 mm					(200mm dia. Sieve)
600	9.50 mm				F2 = _____	MINIMUM REQUIRED TEST
500	6.70 mm					PORTION FOR
400	4.75 mm					INTERMEDIATE GRADING
	Pan 2					MASS OF (400-1200g)
	Washed (W1)					(300mm dia. Sieve)
	Check	(m10) = m8	(m11) = 100	(100) - (m11) = 0		

FINE SIEVING

MAXIMUM MASS	Mass of Split subsample = (m12)			g	**Calculation Details**	MINIMUM REQUIRED TEST
RETAINED EACH	Mass of Adjusted subsample (m12 x W1 / Pan2) +m12 = (m13)			g		PORTION FOR
SIEVE SIZES (mm)	AS.Sieve Size (mm)	Individual mass Retained (m14)	Individual percent (m15)	Total percent Passing	100/(m16) = FACTOR 3	FINE GRADING
150	2.36 mm				(F3) x (m14) = m15	(25 - 150g)
100	1.18 mm					(200mm dia. Sieve)
75	600 um				F3 = _____	
60	425 um					MINIMUM REQUIRED TEST
50	300 um				**Summary**	PORTION FOR
40	150 um					FINE GRADING
25	75 um				GRAVEL =	(50 - 300g)
	Pan 3				SAND =	(300mm dia. Sieve)
	Washed (2) 75um				CLAY & SILT =	
	Check	(m14) = m12 0	(m15) = 100	(100) - (m15) = 0		

NOTE: To obtained adjusted mass M9 = (% passing 26.5 sieve x M8) / 100

To link % mass from (%P4.75) to (%P2.36), M16 = [(M13) x 100/(P4.75)]

Calculation for M_{15} = (F3 x m16)

Tested By: _____ **Checked by:** _____

Date: _____ **Date :** _____

Authorised by: AS(Material Science & Technology Branch)

FORM No.S-3.6.1

Revised Worksheet: 08/04/2022 lhb